

NIRIKSHAN (नरीक्षण) — TAMPER-EVIDENT DRUG TESTING

A Digital Companion for Colorimetric Field Narcotics Testing with Optical Homography, CIELAB Classification, and Cryptographic Custody

e < 0.0012

HOMOGRAPHY ERROR

Sub-pixel ArUco planar rectification

$\Delta E^*00 \leq 0.85$

CIELAB ACCURACY

CIE2000 colorimetric match threshold

100% TAMPER-PROOF

DUAL IMMUTABILITY

SHA-256 sensor raw + HMAC signature

120 kB BUNDLE

TACTICAL UP-PIPE

70% bundle reduction for field 20/30

COMPLETE SYSTEM ARCHITECTURE: 5 INTEGRATED SUBSYSTEMS

SUBSYSTEM 01 : OPTICAL ACQUISITION

REAL-TIME EDGE COMPUTER VISION

- 4-Corner ArUco fiducial lock (DICT_4X4_50)
- Planar homography perspective rectification
- Real-time horizon level & tilt detection (<8°)
- Specular glare saturation rejection (Y > 250)
- Reaction kinetics countdown gate (T_min -> T_max)
- Tactile haptic feedback & auto-shutter lock

Anchor: mobile/app/capture.tex

SUBSYSTEM 02 : ML COLORIMETRIC ENGINE

SPECTRAL DECAY & CIELAB CLASSIFIER

- Retinex Gray-World illuminance normalization
- 18% Neutral Gray card baseline calibration
- Continuous chemical kinetics decay tracking
- CIELAB L*a*b* Euclidean distance ($\Delta E^*00 \leq 0.85$)
- Reagent libraries: Marquis, Mecke, Mandelin, Scott Reagent, Froehde, Simon's, Dille-Koppanyi

Anchor: backend/app/services/classification.py

SUBSYSTEM 03 : CRYPTOGRAPHIC CUSTODY

DUAL-TIER IMMUTABILITY PIPELINE

- Layer 1: SHA-256 raw sensor envelope digest
- Layer 2: HMAC-SHA256 canonical payload seal
- RFC-8785 deterministic JSON key sorting
- Float drift elimination (Lat/Long 6 dec, Acc 2 dec)
- SQLite append-only local cryptographic chain
- Dual-tier offline sync (telemetry vs raw media)

Anchor: backend/app/services/integrity.py

SUBSYSTEM 04 : TACTICAL COMMAND & GEO-INTELLIGENCE

SUPERVISORY AUDIT & NARCOTICS HEATMAPS

- Interactive leaflet map with GPS accuracy radii (±3.2m)
- Real-time officer badge travel velocity anomaly alerts
- Presumptive substance clustering & regional hot-spots
- Strict supervisor reclassification audit trail with reason logs
- High tactile contrast (WCAG AAA) for field command screens
- Zero layout shifts with instant state-driven result filters

Anchor: web/src/pages/dashboard.tex & routes/location.py

SUBSYSTEM 05 : COURT-ADMISSIBLE JUDICIAL DOSSIER

TRIAL-READY EVIDENTIARY EXHIBITS

- Statutory compliance: Sec. 65B Indian Evidence Act / BNSS
- Triad Photographic Exhibit: Raw capture vs Rectified vs Standard
- Air-gapped QR code containing complete signed telemetry
- Instant printable PDF/A certificate via browser @media print
- Reversible privacy redaction for public court records
- Mathematical defense-proof proof: ΔE variance table < 0.85

Anchor: docs/ideas/court-admissible-forensic-export.md

END-TO-END DATAFLOW PIPELINE : FROM FIELD WELL TO JUDICIAL PROOF

